studocu

MAST30027 2020-2 - EXAM

Modern Applied Statistics (University of Melbourne)



Scan to open on Studocu

Downloaded by no no (hahakix253@homvela.com)
Studocu is not sponsored or endorsed by any college or university

THE UNIVERSITY OF MELBOURNE

Student number

Semester 2 Assessment, 2020

School of Mathematics and Statistics

MAST30027 Modern Applied Statistics

Reading time: 30 minutes — Writing time: 2 hours — Upload time: 30 minutes

This exam consists of 15 pages (including this page)

Page 1 of 15 — add any extra pages after page 15 — Page 1 of 15

Permitted Materials

- This exam and/or an offline electronic PDF reader, one or more copies of the masked exam template made available earlier, blank loose-leaf paper and a Casio FX-82 calculator.
- One double sided A4 page of notes (handwritten only).

Instructions to Students

- There are 5 questions with marks as shown. The total number of marks available is 50.
- During writing time you may only interact with the device running the Zoom session with supervisor permission. The screen of any other device must be visible in Zoom from the start of the session.
- If you have a printer, print the exam one-sided. If you cannot print, download the exam to a second device, which must then be disconnected from the internet.
- Write your answers in the boxes provided on the exam that you have printed or the masked exam template that has been previously made available. If you are unable to answer the whole question in the answer space provided then you can append additional handwritten solutions to the end after the 15 numbered pages. If you do this you MUST make a note in the correct answer space or page for the question, warning the marker that you have appended additional remarks at the end.
- If you have been unable to print the exam and do not have the masked template write your answers on A4 paper. The first page should contain only your student number, the subject code and the subject name. Write on one side of each sheet only. Start each question on a new page and include the question number at the top of each page.
- Assemble all exam pages (or masked template pages) in correct page number order and the correct way up. Add any extra pages with additional working at the end. Use a mobile phone scanning application to scan all pages to a single PDF file. Scan from directly above to reduce keystone effects. Check that all pages are clearly readable and cropped to the A4 borders of the original page. Poorly scanned submissions may be impossible to mark.
- Submit your PDF file to the Canvas Assignment corresponding to this exam using the Gradescope window. Before leaving Zoom supervision, confirm with your Zoom supervisor that you have Gradescope confirmation of submission.

©University of Melbourne 2020

Do not place in Baillieu Library
studocu
Downloaded by no no (hahakix253@homvela.com)

Question 1 (7 marks)

Let $X_1, \cdots, X_n$ be independent samples from a Normal distribution $\mathrm{N}(1, \frac{1}{\tau})$ with pdf

$$\sqrt{\frac{\tau}{2\pi}} e^{-\frac{\tau(x-1)^2}{2}}.$$

(a) What is the log-likelihood for this example?

(b) What is the Fisher information for this example?

Page 2 of 15 — add any extra pages after page 15 — Page 2 of 15

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

(c) Find the MLE of τ and its asymptotic distribution.

Page 3 of 15 — add any extra pages after page 15 — Page 3 of 15

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

Page 4 of 15 — add any extra pages after page 15 — Page 4 of 15

Question 2 (9 marks)

The `dvisits` data in the `faraway` package comes from the Australian Health Survey of 1977–78 and consists of 5190 observations on single adults, where young and old have been oversampled. Here, we consider `doctorco` as a response and `sex, age, income, levyplus, freepoor, freerepa, illness, actdays` as predictor variables. The description of each variable is as follows.

- doctorco: number of consultations with a doctor or specialist in the past 2 weeks
- sex: 1 if female, 0 if male
- age: age in years divided by 100
- income: annual income in Australian dollars divided by 1000
- levyplus: 1 if covered by a private health insurance fund for private patients in a public hospital (with doctor of choice), 0 otherwise
- freepoor: 1 if covered by government because of low income, recent immigrant, or unemployed, 0 otherwise
- freerepa: 1 if covered by government because of old-age or disability pension, or because of invalid veteran or family of deceased veteran, 0 otherwise
- illness: number of illnesses in past 2 weeks, with 5 or more coded as 5
- actdays: number of days of reduced activity in past two weeks due to illness or injury

Examine the R code and output below, and then answer the questions that follow.

```
> library(faraway)
> data(dvisits)
> modelA <- glm(doctorco ~ sex + age + income + levyplus + freepoor
+             + freerepa + illness + actdays,
+             family=quasipoisson, data=dvisits)
> summary(modelA)
Call:
glm(formula = doctorco ~ sex + age + income + levyplus + freepoor +
    freerepa + illness + actdays, family = quasipoisson, data = dvisits)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.7696  -0.6865  -0.5773  -0.4906   5.5745

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.055666   0.115808 -17.751   <2e-16 ***
sex          0.163442   0.064454   2.536   0.0112 *
age          0.296311   0.186496   1.589   0.1122
income      -0.195493   0.098511  -1.984   0.0473 *
levyplus     0.143743   0.082153   1.750   0.0802 .
freepoor    -0.404611   0.206938  -1.955   0.0506 .
freerepa     0.118603   0.105656   1.123   0.2617
illness      0.211644   0.019482  10.864   <2e-16 ***
actdays      0.133576   0.005264  25.377   <2e-16 ***
```


studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

Page 5 of 15 — add any extra pages after page 15 — Page 5 of 15

```
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for quasipoisson family taken to be 1.328231)

    Null deviance: 5634.8  on 5189  degrees of freedom
Residual deviance: 4394.3  on 5181  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 6

> modelB <- glm(doctorco ~ sex + age + income,
+             family=quasipoisson, data=dvisits)
> anova(modelB, modelA, test="F")
Analysis of Deviance Table

Model 1: doctorco ~ sex + age + income
Model 2: doctorco ~ sex + age + income + levyplus + freepoor + freerepa +
    illness + actdays
  Resid. Df Resid. Dev Df Deviance      F    Pr(>F)
1      5186     5434.9
2      5181     4394.3  5   1040.5 156.68 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
>
> modelC <- glm(doctorco ~ sex + age + income + levyplus + freepoor
+             + freerepa + illness + actdays,
+             family=poisson, data=dvisits)
> modelD <- glm(doctorco ~ sex + age + income,
+             family=poisson, data=dvisits)
> summary(modelD)

Call:
glm(formula = doctorco ~ sex + age + income, family = poisson,
    data = dvisits)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.0350  -0.8031  -0.6749  -0.6069   6.3695

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.71473    0.09118 -18.805  < 2e-16 ***
sex          0.21565    0.05589   3.859 0.000114 ***
age          1.23798    0.13013   9.514  < 2e-16 ***
income      -0.27726    0.07969  -3.479 0.000502 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)
```


studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

Page 6 of 15 — add any extra pages after page 15 — Page 6 of 15

```
    Null deviance: 5634.8  on 5189  degrees of freedom
Residual deviance: 5434.9  on 5186  degrees of freedom
AIC: 7774.4

Number of Fisher Scoring iterations: 6

> (phiC <- sum(residuals(modelC, type="pearson")^2/modelC$df.residual))
[1] 1.328231
> (phiD <- sum(residuals(modelD, type="pearson")^2/modelD$df.residual))
[1] 2.027811
```

(a) Do you prefer `modelA` or `modelB`, and why?

(b) Show how the F-statistic has been calculated in the analysis of deviance. What are the degrees of freedom for the F statistic?

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

(c) Give the estimator for the coefficient of `illness` and its standard error for `modelC`.

(d) For `modelD`, what is the log-likelihood of the fitted model, and the log-likelihood of the full (saturated) model?

Page 7 of 15 — add any extra pages after page 15 — Page 7 of 15

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

Page 8 of 15 — add any extra pages after page 15 — Page 8 of 15

Question 3 (10 marks)

We assume that the observed data, $X_1 = 1, X_2 = 2, X_3 = 4$ follow a mixture of two Poisson distributions. Specifically, for $i = 1, 2, 3$,

$$Z_i \sim \text{categorical } (\pi, 1 - \pi),$$

$$X_i | Z_i = 1 \sim \text{Poisson}(\lambda_1 = 1.2) \text{ and } X_i | Z_i = 2 \sim \text{Poisson}(\lambda_2 = 2.5),$$

where the Poisson distribution has the probability mass function

$$f(x; \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}.$$

Assume that we derived and implemented the EM algorithm to obtain the MLE of the parameter π.

(a) Assume we ran the EM algorithm two times with different initial values. The following table shows two estimates returned by different runs of the EM algorithm. Which estimate should we use as the MLE of the parameter π? Why?

	initial value	estimate for π
run 1	0.1	0.3
run 2	0.9	0.4

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

Page 9 of 15 — add any extra pages after page 15 — Page 9 of 15

studocu

Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

(b) If $P(Z_i = 1|X_i) > 0.5$, we assign a sample X_i to cluster 1, and cluster 2 otherwise. Using the MLE from the part (a), compute $P(Z_2 = 1|X_2 = 2)$ and $P(Z_2 = 2|X_2 = 2)$. Which cluster is $X_2 = 2$ assigned to?

Page 10 of 15 — add any extra pages after page 15 — Page 10 of 15

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

Page 11 of 15 — add any extra pages after page 15 — Page 11 of 15

Question 4 (18 marks)

Model: We assume that x and y are independent and follow normal distributions:

$$x \sim \mathrm{N}(\mu_1, 1^2) \quad , \quad y \sim \mathrm{N}(\mu_2, 1^2).$$

Prior: We impose the following bivariate normal prior for the mean parameters:

$$\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \text{ with } \boldsymbol{\mu} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ and } \boldsymbol{\Sigma} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix}.$$

Recall that $\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$ and $\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}$, iff $\boldsymbol{x}$ has joint density

$$f_{\boldsymbol{\mu},\boldsymbol{\Sigma}}(\boldsymbol{x}) = \frac{1}{2\pi|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right).$$

(a) We wish to perform posterior inference using the Gibbs sampling. Derive the conditional distribution

$$p(\mu_1|\mu_2, x, y).$$

If it is a known distribution, identify the distribution name and its parameters.

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

(b) We wish to perform posterior inference using the Metropolis-Hastings algorithm. For the current values of the parameters (μ_1^c, μ_2^c), we propose new values (μ_1^n, μ_2^n) as follows: $\mu_1^n \sim \mathrm{N}(0, 1^2)$ and $\mu_2^n \sim \mathrm{N}(\mu_2^c, 1^2)$. Compute the acceptance probability when $(\mu_1^c, \mu_2^c) = (2, 0), (\mu_1^n, \mu_2^n) = (3, 1), x = 1, y = 0$.

Page 12 of 15 — add any extra pages after page 15 — Page 12 of 15

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)

(c) We wish to perform posterior inference using variational inference with the mean-field variational family where $q(\mu_1, \mu_2) = q_1(\mu_1)q_2(\mu_2)$ and use the CAVI algorithm for optimisation. The CAVI algorithm iteratively optimises each factor as follows while holding the other factor fixed:

$$\begin{aligned} q_1^*(\mu_1) &\propto \exp\{\mathbb{E}_{\mu_2}[\log p(\mu_1, \mu_2, x, y)]\}, \\ q_2^*(\mu_2) &\propto \exp\{\mathbb{E}_{\mu_1}[\log p(\mu_1, \mu_2, x, y)]\}, \end{aligned}$$

where the expectations $\mathbb{E}_{\mu_2}$ and $\mathbb{E}_{\mu_1}$ are taken with respect to $q_2^*(\mu_2)$ and $q_1^*(\mu_1)$, respectively. Derive $q_1^*(\mu_1)$ and $q_2^*(\mu_2)$, and identify the corresponding distribution names and their parameters.

Page 13 of 15 — add any extra pages after page 15 — Page 13 of 15

studocu
Scan to open on Studocu
Downloaded by no no (hahakix253@homvela.com)